PHYS4055 Week 4 Tutorial Solutions

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Tutorial date: 20 August 2026

Question 1: Entanglement distillation

Alice and Bob share n copies of the mixed state

$$\rho_{AB} = p\left|\phi^+\right\rangle\left\langle\phi^+\right| + (1-p)\left|\psi^+\right\rangle\left\langle\psi^+\right|, \qquad p > \frac{1}{2},$$

where

$$\left|\phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right), \qquad \left|\psi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right)$$

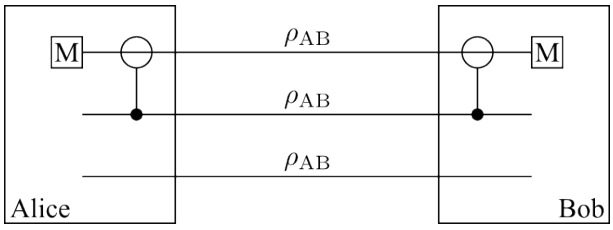
are Bell states. The goal of entanglement distillation is to use several imperfectly entangled pairs to obtain fewer pairs with higher entanglement.

For one step of the protocol, Alice and Bob take two copies of ρ_{AB}. On each side, they apply a CNOT gate, using the qubit from the first pair as the control qubit and the qubit from the second pair as the target qubit. The unitary operation of the CNOT gate can be written as

$$\hat{U}_{\text{CNOT}} = |0_C\rangle\langle 0_C| \otimes \hat{I}_T + |1_C\rangle\langle 1_C| \otimes \hat{\sigma}_{x,T},$$

that is, the target qubit is flipped if and only if the control qubit is $|1\rangle$.

After applying the two CNOT gates, Alice and Bob measure the target pair in the computational basis $\{|0\rangle, |1\rangle\}$ as shown in the figure below (where the first line is the target and the second line is the control). If they obtain the same measurement outcome, they keep the control pair; otherwise, they discard it.



(a) Calculate the state after one successful step:

  (a.1) Calculate the initial joint system.

  (a.2) Apply the CNOT gates.

  (a.3) Measure in the described basis and determine the resulting state.

(b) What is the probability that one distillation step is successful?

(c) Calculate the fidelity of the resulting state with respect to $|\phi^+\rangle$. Is it larger or smaller than the initial fidelity?

(d) Suppose that, after one successful step, Alice and Bob repeat the same procedure using pairs with the improved fidelity. What do you expect to happen to the fidelity after several successful rounds? What is the trade-off?

Solution

(a.1) We label the two copies as pair 1 and pair 2, which correspond to control and target pairs, respectively. The two-copy state is

$$\begin{aligned}\rho_{AB}^{\otimes 2} = p^2 \left|\phi^+\right\rangle_1 \left|\phi^+\right\rangle_2 \left\langle\phi^+\right|_1 \left\langle\phi^+\right|_2 \\ + p(1-p) \left|\phi^+\right\rangle_1 \left|\psi^+\right\rangle_2 \left\langle\phi^+\right|_1 \left\langle\psi^+\right|_2 \\ + p(1-p) \left|\psi^+\right\rangle_1 \left|\phi^+\right\rangle_2 \left\langle\psi^+\right|_1 \left\langle\phi^+\right|_2 \\ + (1-p)^2 \left|\psi^+\right\rangle_1 \left|\psi^+\right\rangle_2 \left\langle\psi^+\right|_1 \left\langle\psi^+\right|_2 .\end{aligned}$$

This is a mixed state that yields the four possible two-pair input states with probabilities

input state	probability
$\left\|\phi^+\right\rangle_1 \left\|\phi^+\right\rangle_2$	p^2
$\left\|\phi^+\right\rangle_1 \left\|\psi^+\right\rangle_2$	$p(1-p)$
$\left\|\psi^+\right\rangle_1 \left\|\phi^+\right\rangle_2$	$p(1-p)$
$\left\|\psi^+\right\rangle_1 \left\|\psi^+\right\rangle_2$	$(1-p)^2$

(a.2) Now Alice applies a CNOT from her qubit in pair 1 (control) to her qubit in pair 2 (target), and Bob does the same on his side. We can track separately what happens to the four possible Bell-state combinations, keeping in mind that Alice operates on the first qubit of each Bell state, and Bob on the second qubit. If the control pair is $|a_1 b_1\rangle_1$ and the target pair is $|a_2 b_2\rangle_2$, the bilateral CNOT acts as

$$|a_1 b_1\rangle_1 |a_2 b_2\rangle_2 \longmapsto |a_1 b_1\rangle_1 |a_1 \oplus a_2, b_1 \oplus b_2\rangle_2 .$$

We now apply this to the four possible input states:

$$\begin{aligned}\hat{U}_{\text{CNOT}} \left|\phi^+\right\rangle_1 \left|\phi^+\right\rangle_2 &= \hat{U}_{\text{CNOT}} \frac{1}{2} \left(|00\rangle_1 |00\rangle_2 + |00\rangle_1 |11\rangle_2 + |11\rangle_1 |00\rangle_2 + |11\rangle_1 |11\rangle_2\right) \\ &= \frac{1}{2} \left(|00\rangle_1 |00\rangle_2 + |00\rangle_1 |11\rangle_2 + |11\rangle_1 |11\rangle_2 + |11\rangle_1 |00\rangle_2\right) \\ &= \left|\phi^+\right\rangle_1 \left|\phi^+\right\rangle_2 ,\end{aligned}$$

and similarly

$$\begin{aligned}\hat{U}_{\text{CNOT}} \left|\phi^+\right\rangle_1 \left|\psi^+\right\rangle_2 &= \left|\phi^+\right\rangle_1 \left|\psi^+\right\rangle_2 , \\ \hat{U}_{\text{CNOT}} \left|\psi^+\right\rangle_1 \left|\phi^+\right\rangle_2 &= \left|\psi^+\right\rangle_1 \left|\psi^+\right\rangle_2 , \\ \hat{U}_{\text{CNOT}} \left|\psi^+\right\rangle_1 \left|\psi^+\right\rangle_2 &= \left|\psi^+\right\rangle_1 \left|\phi^+\right\rangle_2 .\end{aligned}$$

(a.3) Alice and Bob measure the target pair in the computational basis. A successful event corresponds to Alice and Bob obtaining the same result, i.e., the target pair is projected onto either $|00\rangle_2$ or $|11\rangle_2$. The corresponding projection operator is

$$\Pi_{\text{s}} = \hat{I}_1 \otimes \left(|00\rangle_2 \langle 00|_2 + |11\rangle_2 \langle 11|_2\right) .$$

Applying this projection to the four cases after the CNOT gives

$$\begin{aligned}\Pi_{\text{s}} \left|\phi^+\right\rangle_1 \left|\phi^+\right\rangle_2 &= \left|\phi^+\right\rangle_1 \left|\phi^+\right\rangle_2 , \\ \Pi_{\text{s}} \left|\phi^+\right\rangle_1 \left|\psi^+\right\rangle_2 &= 0 , \\ \Pi_{\text{s}} \left|\psi^+\right\rangle_1 \left|\psi^+\right\rangle_2 &= 0 , \\ \Pi_{\text{s}} \left|\psi^+\right\rangle_1 \left|\phi^+\right\rangle_2 &= \left|\psi^+\right\rangle_1 \left|\phi^+\right\rangle_2 .\end{aligned}$$

After the measurement, the target pair is discarded. Therefore, the surviving control-pair states are $|\phi^+\rangle_1$ and $|\psi^+\rangle_1$. Since the surviving contributions came from the initial components $|\phi^+\rangle_1 |\phi^+\rangle_2$ and $|\psi^+\rangle_1 |\psi^+\rangle_2$, with probabilities p^2 and $(1-p)^2$, respectively, the unnormalised state of the retained pair is

$$\rho'_{\text{unnorm}} = p^2 \left|\phi^+\right\rangle \left\langle\phi^+\right| + (1-p)^2 \left|\psi^+\right\rangle \left\langle\psi^+\right|.$$

Normalising, the state after one successful step is

$$\rho'_{AB} = \frac{p^2 |\phi^+\rangle \langle\phi^+| + (1-p)^2 |\psi^+\rangle \langle\psi^+|}{p^2 + (1-p)^2}.$$

(b) The probability of success is given by the coefficients of the unnormalised output state:

$$P_{\text{s}} = p^2 + (1-p)^2.$$

(c) The fidelity with respect to $|\phi^+\rangle$ is

$$F' = \left\langle\phi^+\right| \rho'_{AB} \left|\phi^+\right\rangle = F' = \frac{p^2}{p^2 + (1-p)^2}.$$

Since $p > 1/2$, this value is greater than the initial fidelity $F = p$. Therefore, one successful distillation step increases the fidelity with respect to $|\phi^+\rangle$.

(d) If Alice and Bob repeat the protocol using two pairs with the same current fidelity p_k, then after a successful step the fidelity becomes

$$p_{k+1} = \frac{p_k^2}{p_k^2 + (1-p_k)^2}.$$

Since in the initial state $p_0 > 1/2$, the sequence increases towards 1. Thus, the retained pairs become closer and closer to the maximally entangled Bell state $|\phi^+\rangle$.

The trade-off is that the protocol consumes entangled pairs for measurements and only keeps the result when the measurement outcomes agree, which happens probabilistically. Therefore, the fidelity of the retained pairs increases, but the total number of remaining pairs decreases.